HR ANALYTICS DASHBOARD

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1. INTRODUCTION

Human Resource Management plays an important role in understanding employee behavior, satisfaction, performance, and retention. Organizations generate large amounts of employee-related data, but extracting meaningful insights from this data can be challenging without proper analytical techniques.
The HR Analytics project focuses on analyzing employee information to understand workforce characteristics and identify patterns related to employee attrition. The project uses data cleaning, exploratory analysis, SQL-based analysis, and interactive Power BI visualizations to transform raw HR data into meaningful business insights.
The analysis covers employee demographics, departments, job roles, salary, experience, satisfaction, performance, workload, overtime, and attrition. The final Power BI dashboard provides an interactive view of important HR metrics and helps users explore employee and attrition patterns.

2. PROBLEM STATEMENT

Organizations need to continuously monitor employee-related factors to understand workforce trends and reduce employee turnover. However, raw employee data contains multiple attributes such as age, department, job role, income, satisfaction, performance, overtime, and experience, making it difficult to identify important patterns manually.

The problem addressed in this project is to analyze employee data and identify meaningful patterns related to employee attrition. The project aims to determine how attrition varies across departments, job roles, overtime status, marital status, and performance ratings, while also providing an overall view of the organization's workforce.

3. OBJECTIVE

- To analyze the overall employee workforce.
- To understand employee distribution across departments and job roles.
- To analyze employee salary and experience patterns.
- To calculate important HR metrics such as total employees and attrition rate.
- To identify patterns in employee attrition.
- To compare attrition across departments and job roles.
- To analyze the relationship between overtime and attrition.
- To analyze attrition based on marital status and performance rating.
- To create an interactive HR analytics dashboard.
- To present meaningful insights that can support HR decision-making.

4. Dataset Description

- **Dataset Overview**

  The HR Analytics dataset contains information about 10,000 employees and 25 attributes. The dataset includes demographic, job-related, salary, experience, satisfaction, performance, workload, and attrition-related information.
  The main target variable in the dataset is **Attrition**, which indicates whether an employee has left the organization. The value **"Yes"** represents employees who left, while **"No"** represents employees who stayed.

- **Dataset Features**

Category	Features
Employee Details	Employee_ID, Age, Gender, Marital_Status
Job Details	Department, Job_Role, Job_Level
Salary	Monthly_Income, Hourly_Rate
Experience	Years_at_Company, Years_in_Current_Role, Years_Since_Last_Promotion
Satisfaction	Work_Life_Balance, Job_Satisfaction, Work_Environment_Satisfaction, Relationship_with_Manager
Performance	Performance_Rating, Job_Involvement
Workload	Overtime, Project_Count, Average_Hours_Worked_per_Week, Absenteeism
Career/Training	Training_Hours_Last_Year, Number_of_Companies_Worked
Target	Attrition

5. Tools & Technologies

Tool	Purpose
Python	Data cleaning and exploratory analysis
Pandas	Data manipulation and transformation
Matplotlib	Data visualization

Tool	Purpose
PostgreSQL	SQL-based data analysis
Power BI	Interactive dashboard and visualization
Microsoft Word	Project documentation

6. Data Cleaning & Preparation

- **Data Inspection**
  The dataset was initially inspected to understand its structure, number of records, column names, and data types. Python and Pandas were used to examine the dataset and identify potential data-quality issues.
- **Data Cleaning**
  Dataset structure checked

  Data types verified

  Missing values checked

  Duplicate records checked

  Column names reviewed

  Data consistency checked

- **Age Group Creation**
  To improve employee age analysis, a new categorical variable called **Age_Group** was created by grouping employees into different age ranges.
- Groups:

  18–25

  26–35

  36–45

  46–55

  56–65

- **Data Preparation**
- After cleaning and transformation, the prepared dataset was used for SQL analysis and Power BI visualization. The cleaned data ensured consistency across the analytical workflow.

7. SQL Analysis

- **Basic Employee Analysis**

  Total number of employees

  Employees who left

  Employees based on age

  Employees by department

  Employees by job role

- **Salary Analysis**

  Average monthly income

  Average income by department

  Average income by job role

  Highest salary analysis

- **Attrition Analysis**

  Total attrition

  Attrition rate

  Attrition by department

  Attrition by job role

  Attrition by overtime

  Attrition by marital status

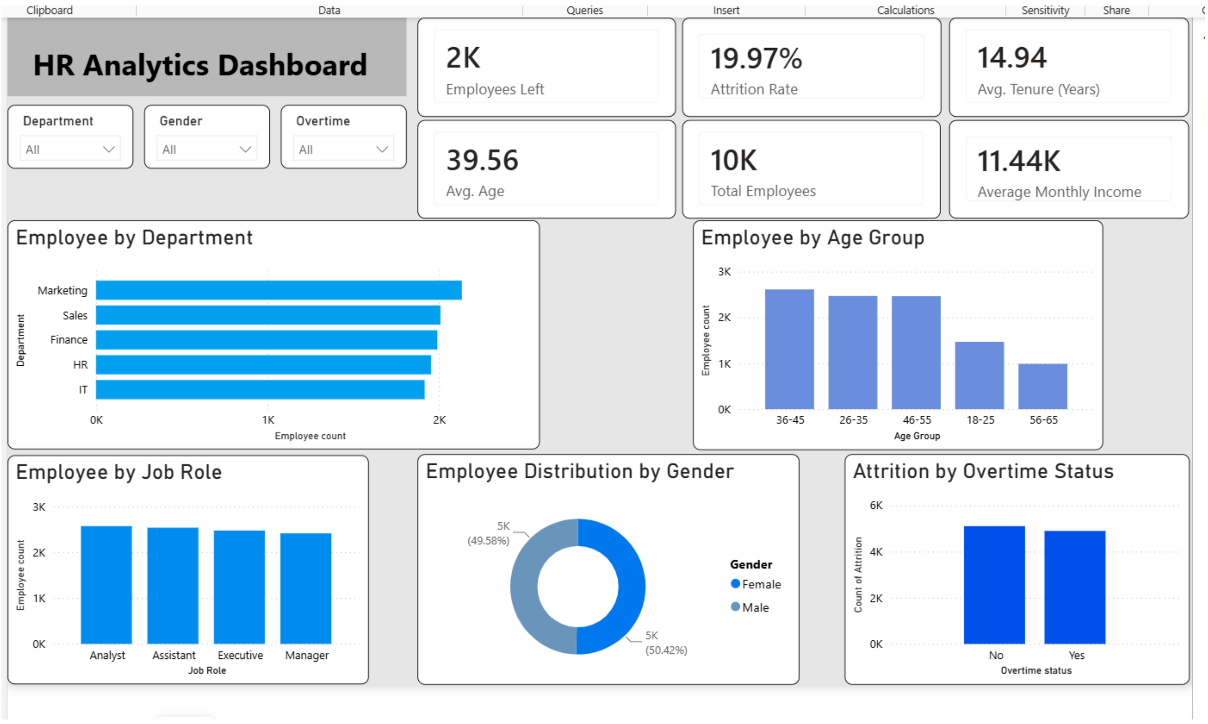
  Attrition by performance rating

- **Example SQL**
-
```
-- Find attrition rate by department
SELECT
Department,
ROUND(
COUNT(*) FILTER (WHERE Attrition = 'Yes') * 100.0 / COUNT(*),
2
) AS attrition_rate
FROM HR
GROUP BY Department
ORDER BY attrition_rate DESC;
```

- **Explanation:**
  This query calculates the percentage of employees who left the organization in each department. It helps identify departments with relatively higher attrition rates.

8. Power BI Dashboard

- **Dashboard Page 1 — Workforce Overview**



- Page 1 contains:
- **KPI Cards**

  Total Employees

  Employees Left

  Attrition Rate

  Avg. Tenure

  Avg. Age

  Avg. Monthly Income

- **Slicers**

  Department

Gender

Overtime

- **Visualizations**

Employees by Department

Employees by Age Group

Employees by Job Role

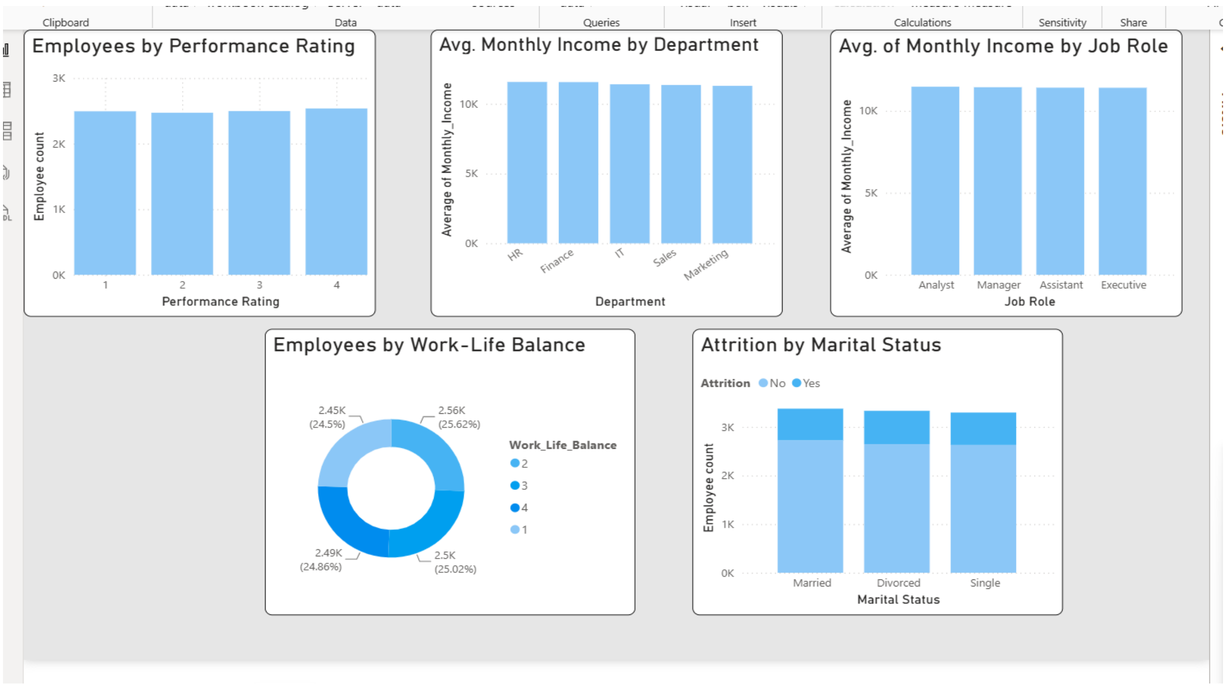
Employees by Gender

Attrition by Overtime

- **Purpose**

Page 1 provides an overall overview of the organization's workforce and key HR metrics. Users can interact with slicers to analyze employee data based on department, gender, and overtime status

- **Dashboard Page 2 — Detailed HR Analysis**



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- Page 2 contains:

Employees by Performance Rating

Avg. Monthly Income by Department

Avg. Monthly Income by Job Role

Employees by Work-Life Balance

Attrition by Marital Status

- **Purpose**
  Page 2 provides a more detailed analysis of employee performance, income, work-life balance, and attrition patterns. It allows users to explore different aspects of employee behavior and organizational workforce characteristics.

9. Key Insights & Findings

- Finding 1 — Overall Attrition

  The organization has **10,000 employees**, of which **1,997 employees have left**, resulting in an overall attrition rate of **19.97%**.

- Finding 2 — Department Attrition

  Finance has the highest attrition rate at **20.85%**, followed by IT at **20.35%**. Marketing has the lowest attrition rate at **19.36%**.

- Finding 3 — Job Role Attrition

  Assistant roles have the highest attrition rate at **21.43%**, while Manager roles have the lowest attrition rate at **18.06%**.

- Finding 4 — Overtime

  Employees working overtime have a slightly higher attrition rate of **20.07%**, compared with **19.87%** for employees who do not work overtime.

- Finding 5 — Marital Status

  Divorced employees have the highest attrition rate at **20.54%**, while married employees have the lowest at **19.23%**.

- Finding 6 — Performance Rating

  Employees with a performance rating of **3** have the highest attrition rate at **21.23%**, while employees with a rating of **1** have the lowest at **19.29%**.

- Overall Finding

  Overall, attrition rates across the analyzed categories are relatively close, indicating that employee turnover is not dominated by a single factor. However, certain groups such as Assistant roles and the Finance department show comparatively higher attrition rates and may require further investigation.

10. Conclusion

- The HR Analytics project successfully transformed raw employee data into meaningful analytical insights using Python, PostgreSQL, and Power BI. The data was cleaned and prepared before performing SQL-based analysis and creating interactive visualizations.
- The analysis provided an overview of employee demographics, job roles, income, experience, satisfaction, performance, and attrition. The Power BI dashboard presents important HR metrics through interactive KPIs, charts, and slicers, allowing users to explore workforce patterns efficiently.
- The analysis identified differences in attrition across departments, job roles, overtime status, marital status, and performance ratings. Although the differences between several groups are relatively small, the findings can help HR teams identify areas that may require deeper investigation.
- Overall, the project demonstrates how data analytics can be used to convert employee data into actionable information for better workforce monitoring and HR decision-making.

11. Future Scope

- Develop predictive models to identify employees at high risk of attrition.
- Use machine learning techniques for employee attrition prediction.
- Add real-time HR data integration.
- Include additional employee engagement metrics.
- Analyze salary growth and promotion patterns.
- Perform deeper analysis of employee satisfaction.
- Develop automated Power BI reports.
- Add advanced statistical analysis to identify significant attrition factors.
- Compare attrition trends over multiple time periods.
- Integrate the dashboard with organizational HR systems.